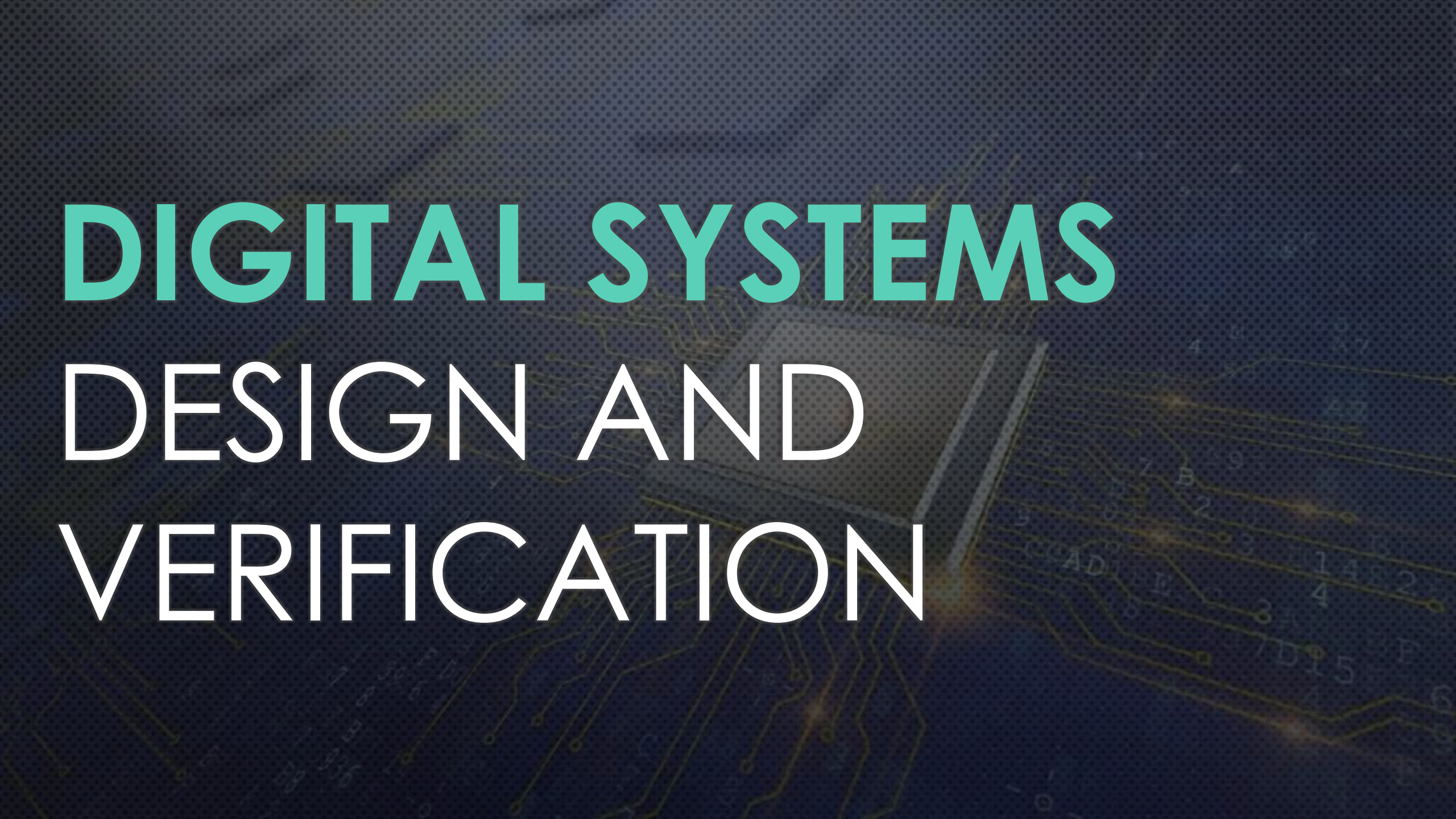


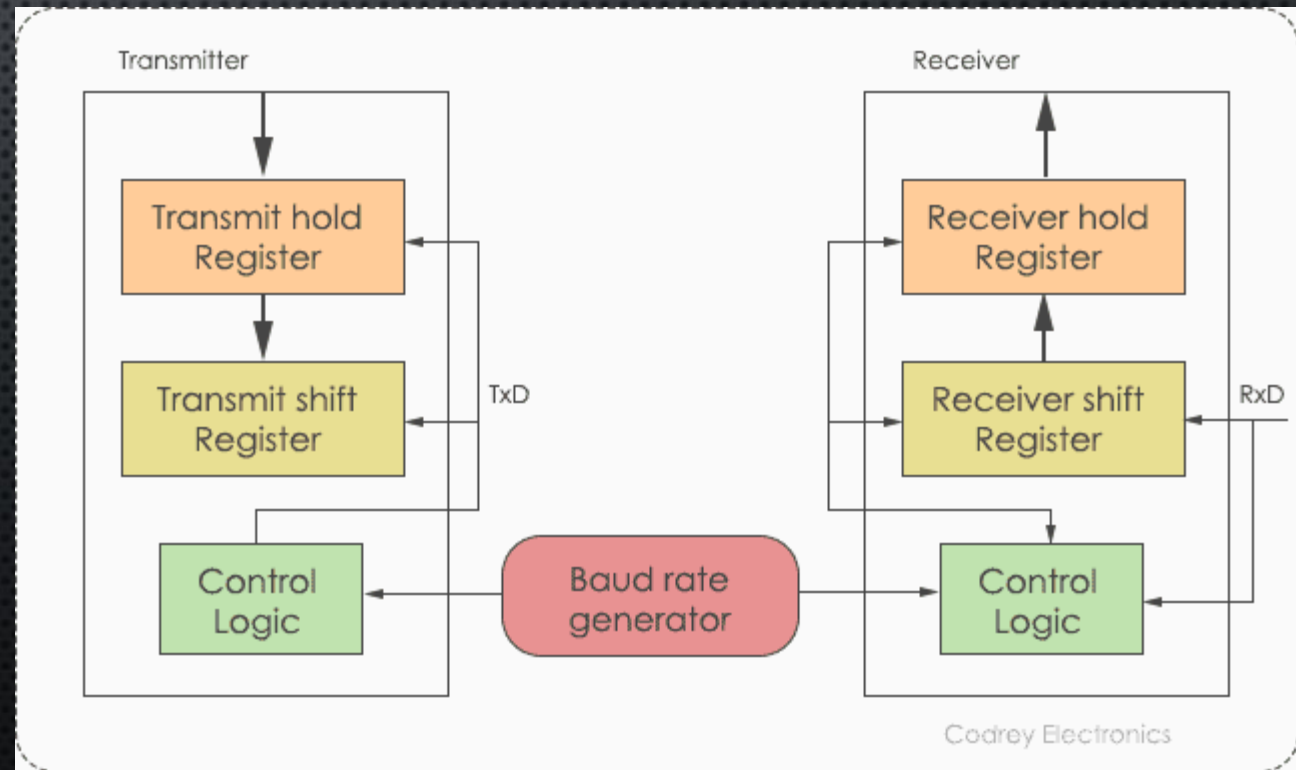
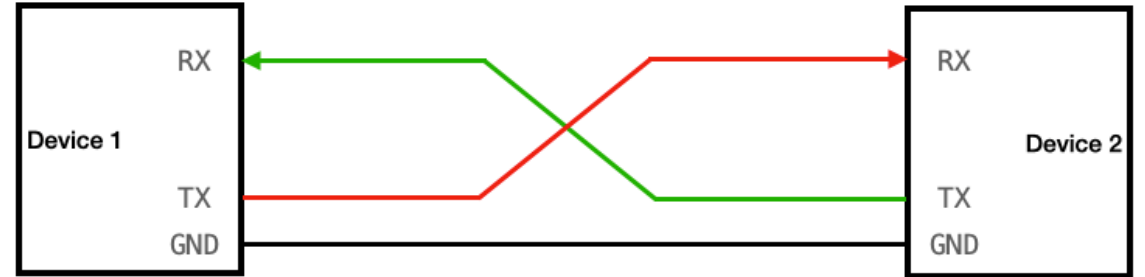
The background of the slide is a dark blue, semi-transparent image of a printed circuit board (PCB) with intricate gold-colored traces. A fine, light-blue grid is overlaid on the entire image. The text is centered and consists of three lines: 'DIGITAL SYSTEMS' in a bold, teal, sans-serif font, and 'DESIGN AND VERIFICATION' in a bold, white, sans-serif font.

DIGITAL SYSTEMS DESIGN AND VERIFICATION

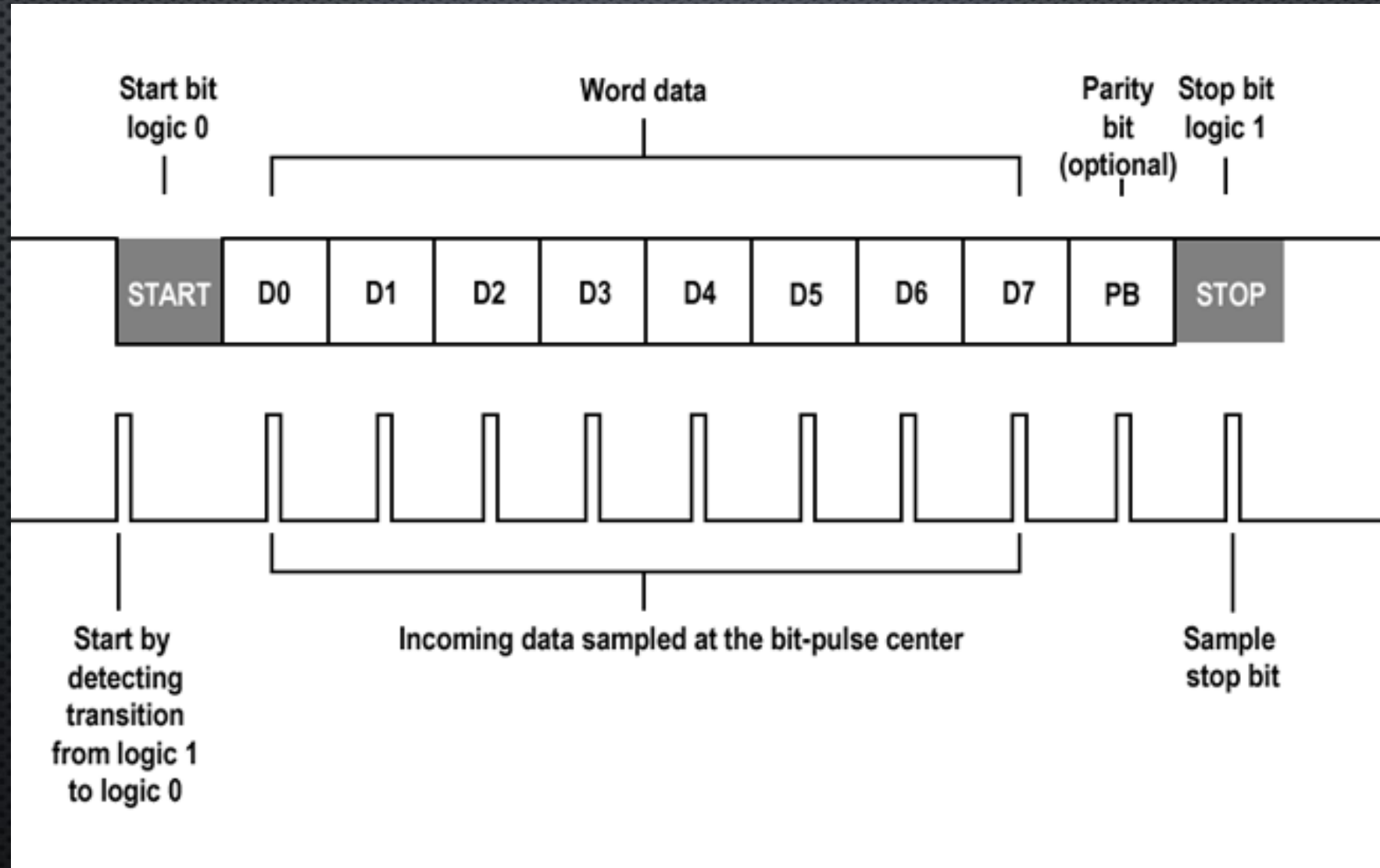
UART

Characteristics:

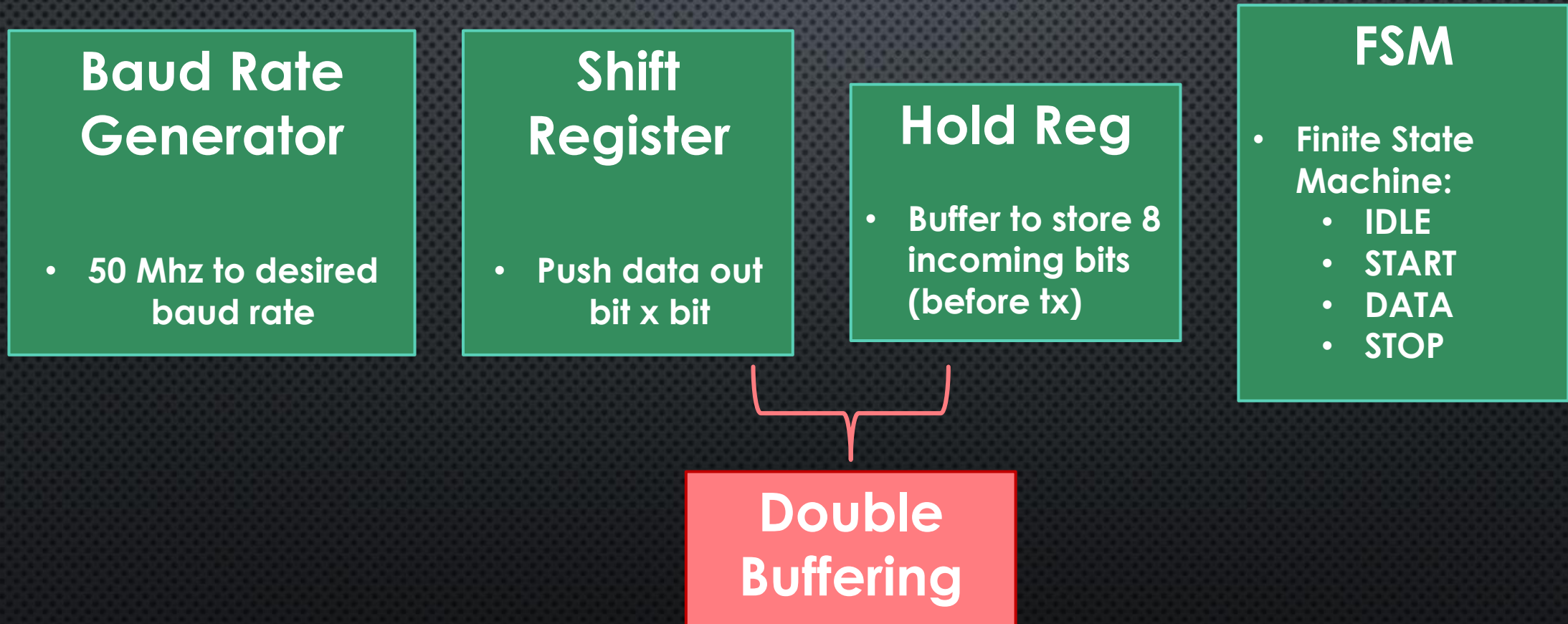
- **Asynchronous:** No shared clock signal between devices
- **Wires:** 2 (TX and RX) + Ground
- Devices *MUST* agree on Baud Rate beforehand.
- Full-Duplex



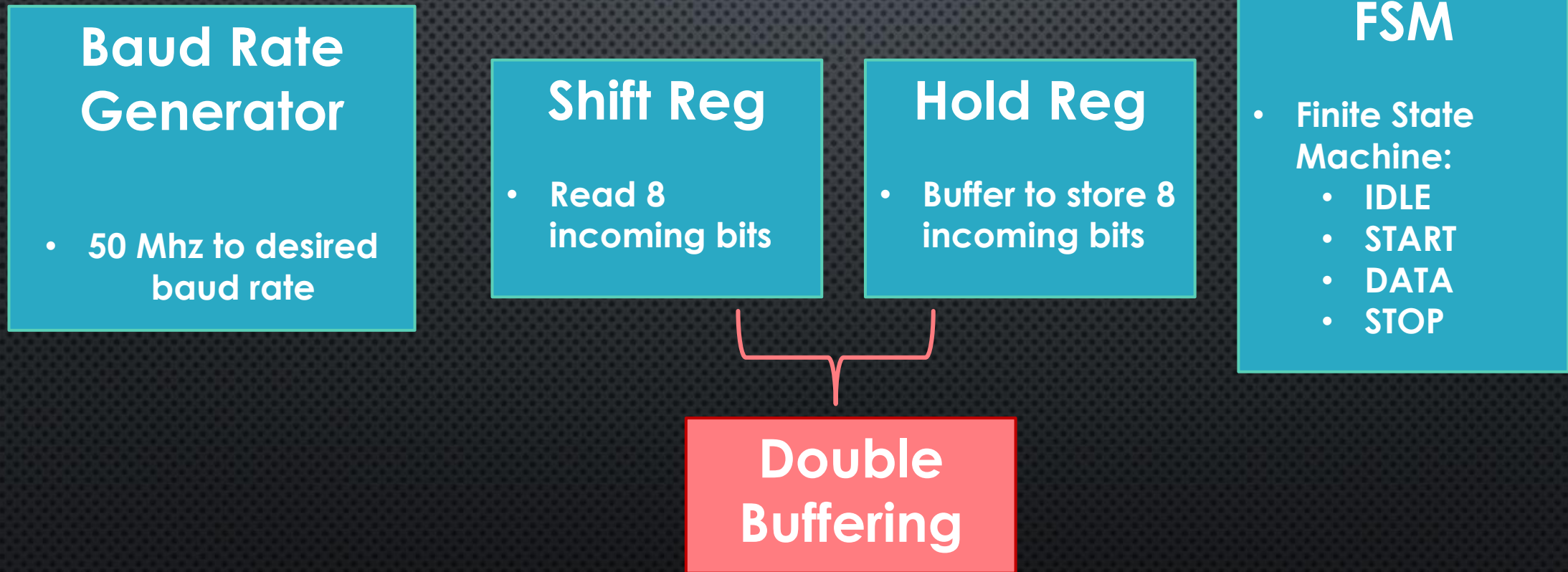
UART - FRAME



UART - TRANSMITTER



UART - RECEIVER



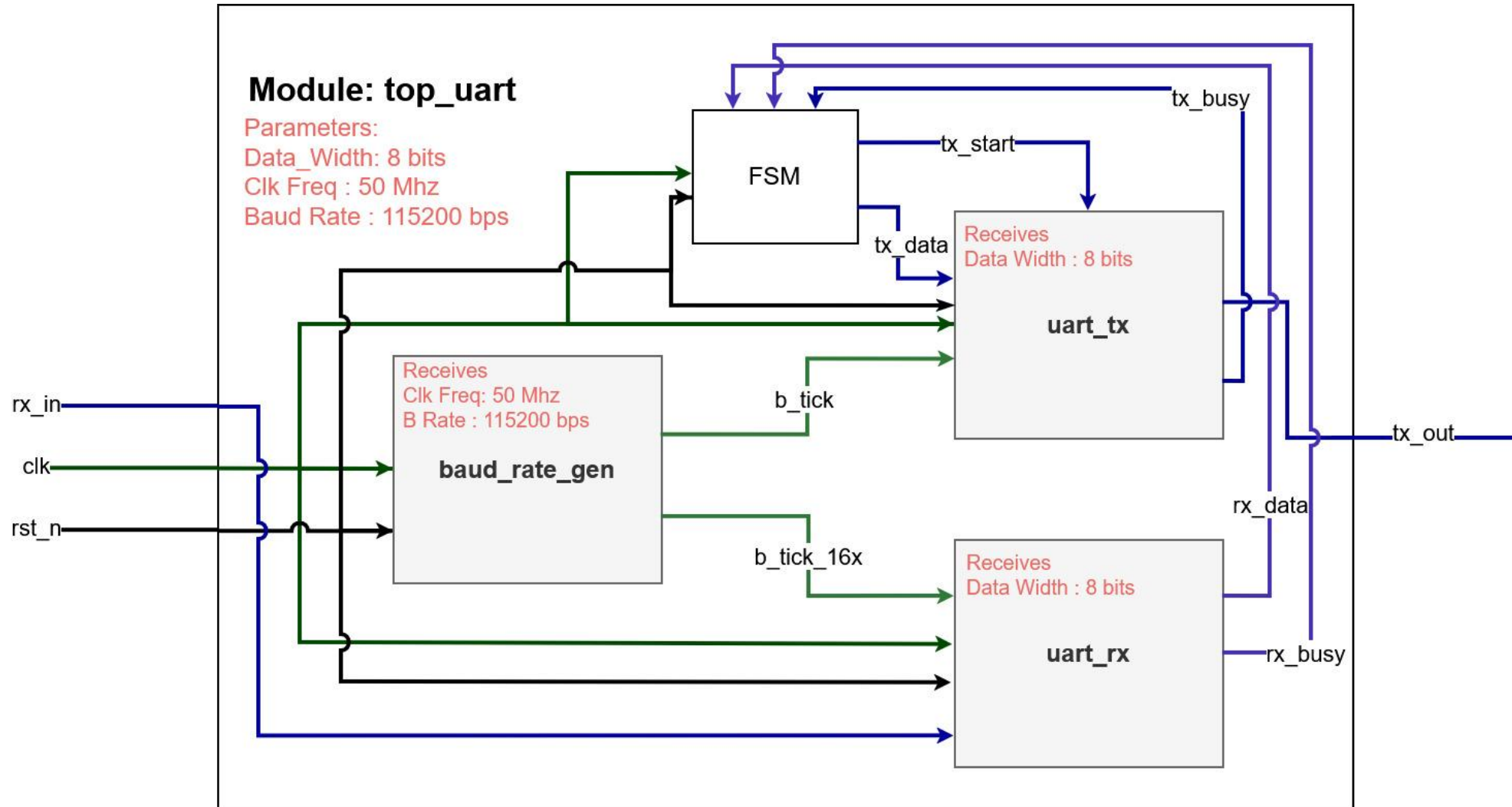
RECEIVER ARCHITECTURE

Capturing asynchronous data requires a robust digital chain to ensure signal integrity:

- **16x Oversampling:** Synchronizing a clock-less signal.
- **RX Shift Register:** Converting temporal serial bits to spatial parallel bytes.
- **RX Hold Register:** Providing data persistence for the Controller (MCU).

- By sampling 16 times faster than the baud rate, we can filter out glitches and transients that are shorter than a full bit period.
- We wait 8 ticks after a START edge to sample exactly in the middle of the bit, where the voltage is most stable.

FULL SYSTEM BLOCK DIAGRAM



WHY GIT ?



Code_v1



Code_v1final



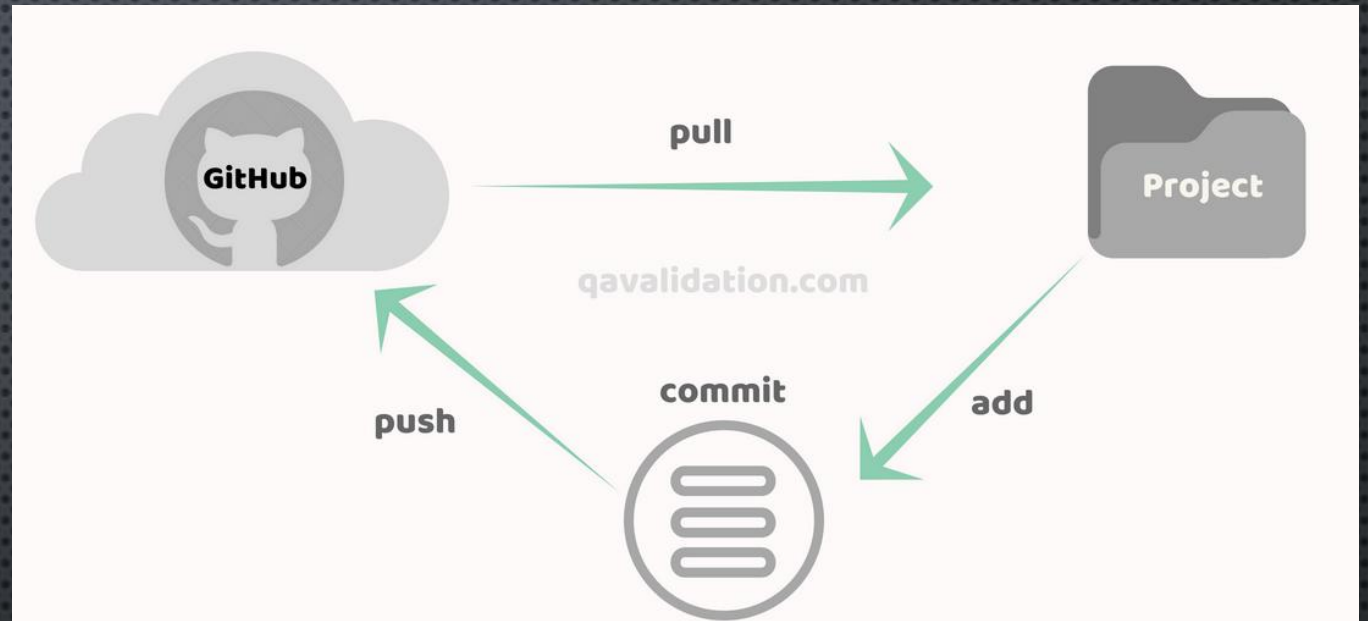
Code_FINAL



Code_v1final_final

GIT BASICS

- Git is LOCAL
- Git HUB is a backup + collaboration hub
- Every commit is a frozen state, checkpoint, snapshot, etc.
 - Of the entire project
- You can always go back to any previous commit



Key bash commands:

git **add** file.v # Stage changes

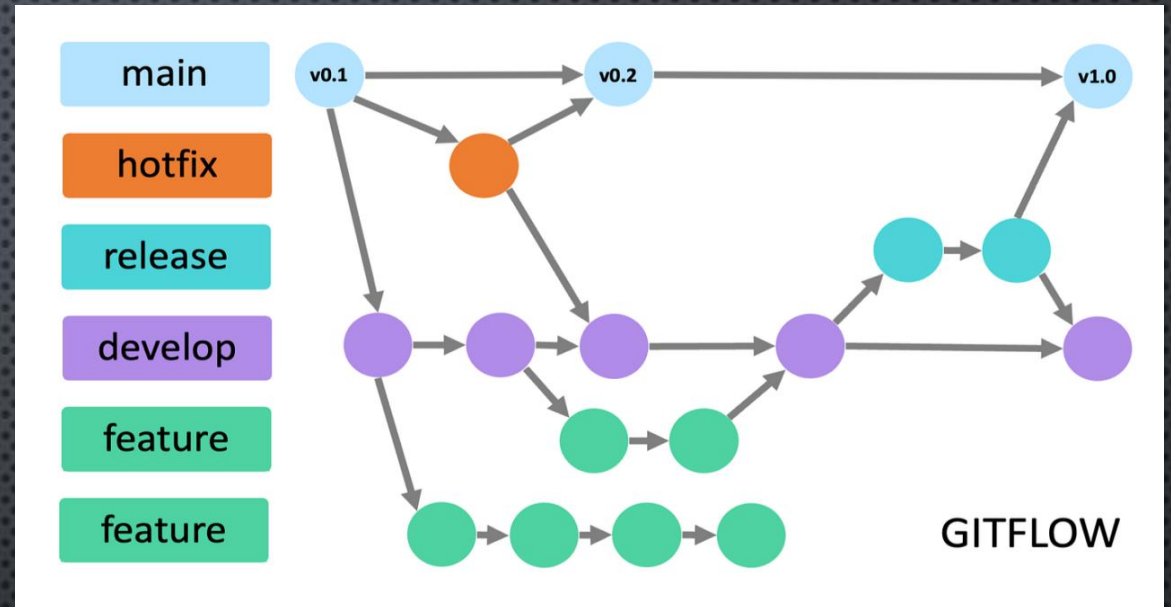
git **commit** -m "Add MUX" # Commit to local repo

git **push** # Send to GitHub

git **pull** # Get latest from GitHub

BRANCHES

- **Branch** = Separate line of development
- Test without breaking the main code
- When ready, merge back to main
- It is suggested to create a feature/ branch for each new feature



Key bash commands:

```
git branch feature/mux # Create new branch
git checkout feature/mux # Switch to branch
git checkout main      # Go back to main
git merge feature/mux  # Merge the branch
```

FINAL EXERCISE !!!